



PRESIDENCY UNIVERSITY

(Established under the Presidency University Act, 2013 of the Karnataka Act 41 of 2013)

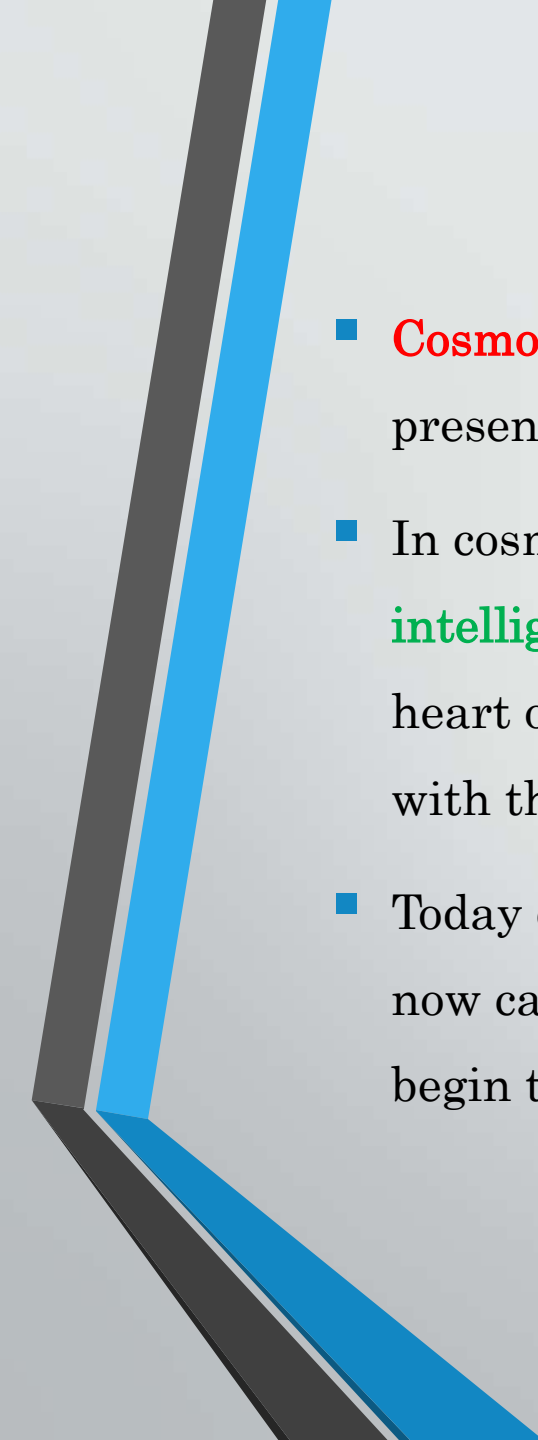
Astrophysics

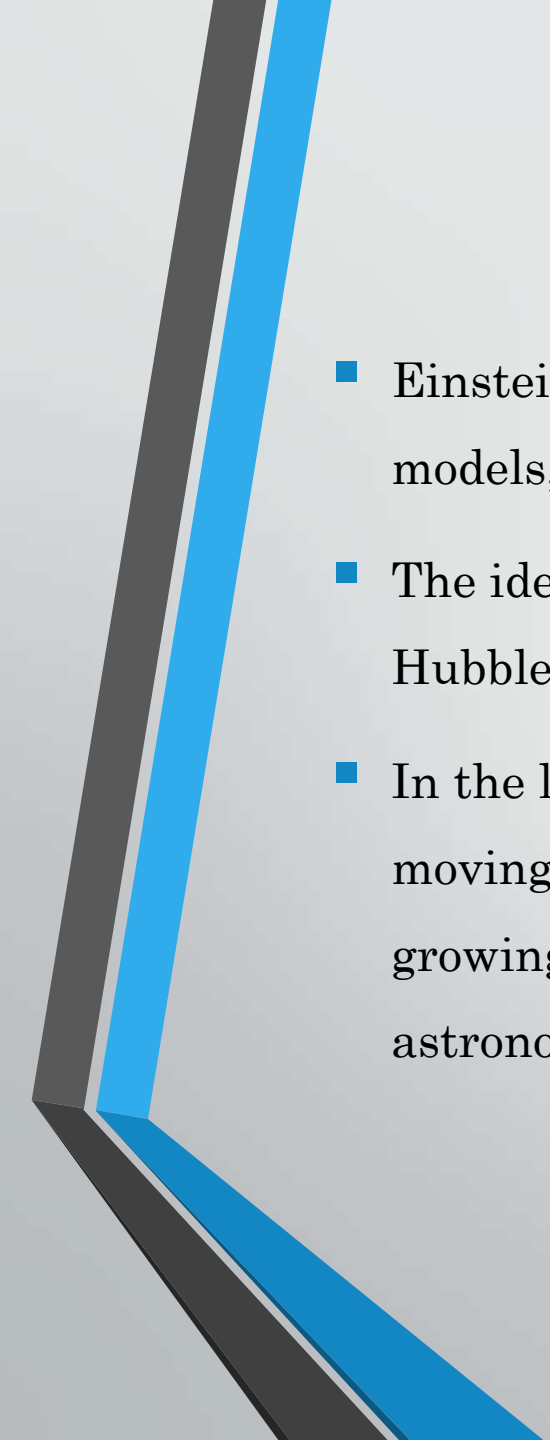
Dr. Bharathi D
SOE-Physics Department



Space:

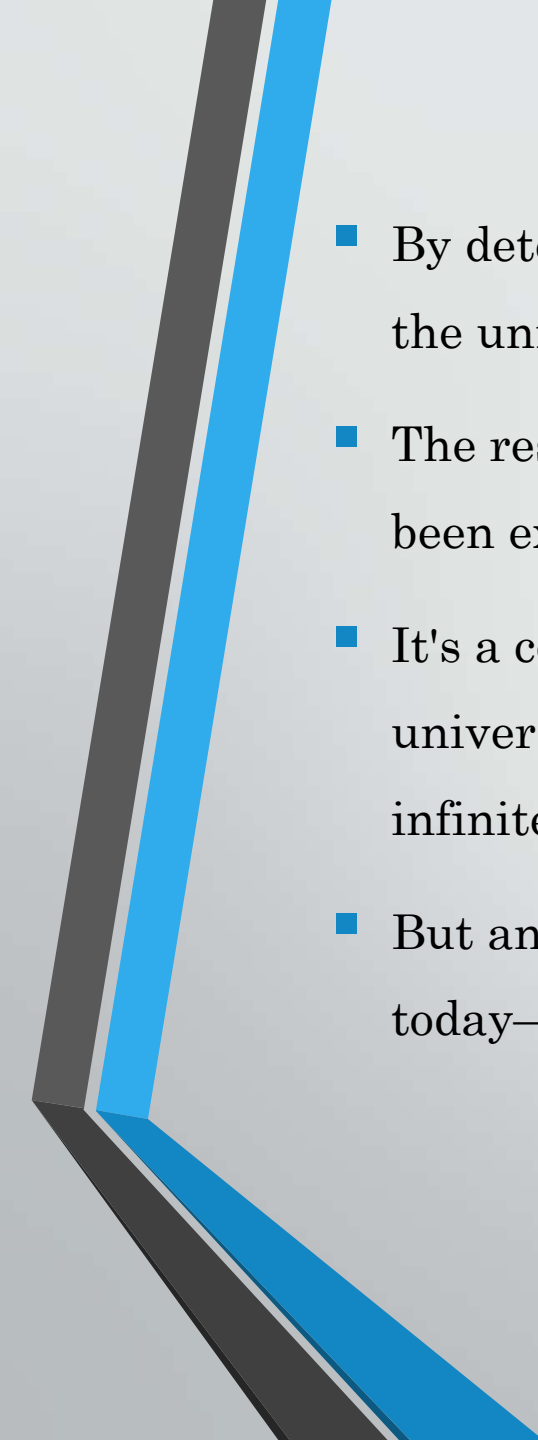
- Space means everything that occupies between planets, moons, stars, etc.
- In simple words space is everything beyond the top of Earth's atmosphere- the satellites, Moon, mars, Milky way, Galaxies and much more.
- To know more about the space astronomy, the galaxies, satellites, asteroids, planets and other things in the space, it is not possible for every individual to go there and explore the galaxies and the universe. This is why we have created an equivalent tool that will aid you in exploring more about universe and its constituents.

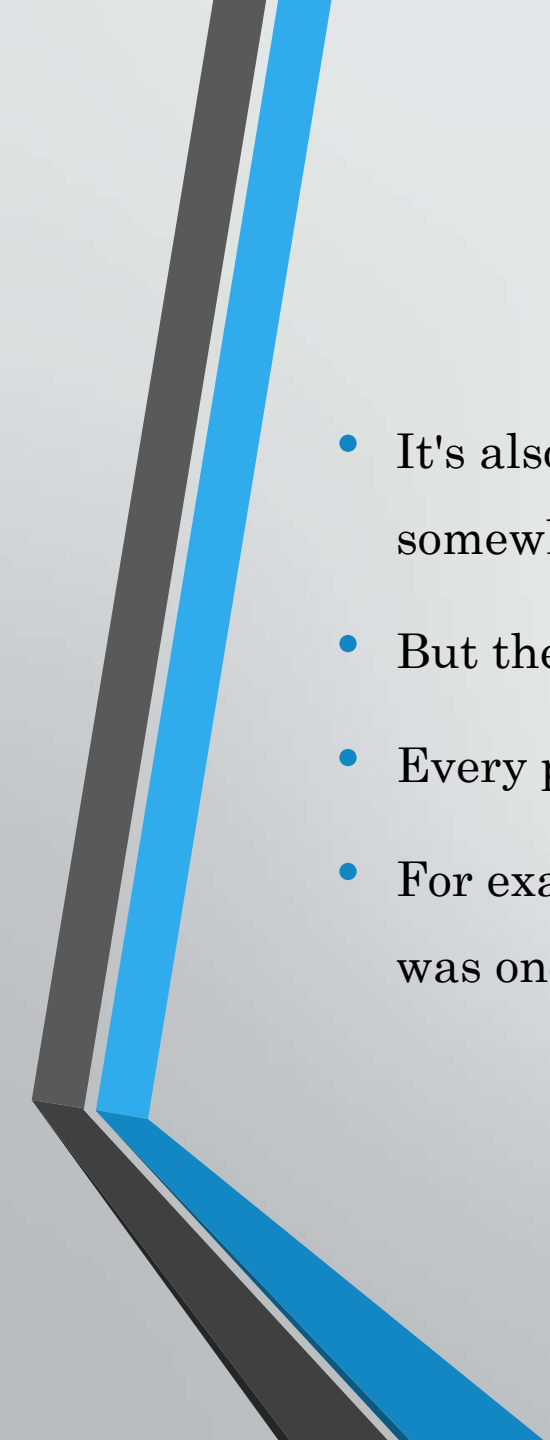
- 
- **Cosmology** is the study of the evolution of the universe from its first moments to the present.
 - In cosmology the most fundamental question we can ask is: **Does our universe have intelligible regularities that we can understand—is it ordered?** This question lies at the heart of the scientific revolution beginning in the sixteenth century. That revolution began with the discoveries by Copernicus, Galileo, and Newton of order in our world.
 - Today our scientific understanding of nature's order has reached a critical threshold. Only now can we begin to piece together a coherent picture of the whole. Only now can we begin to see the deep order of our universe.

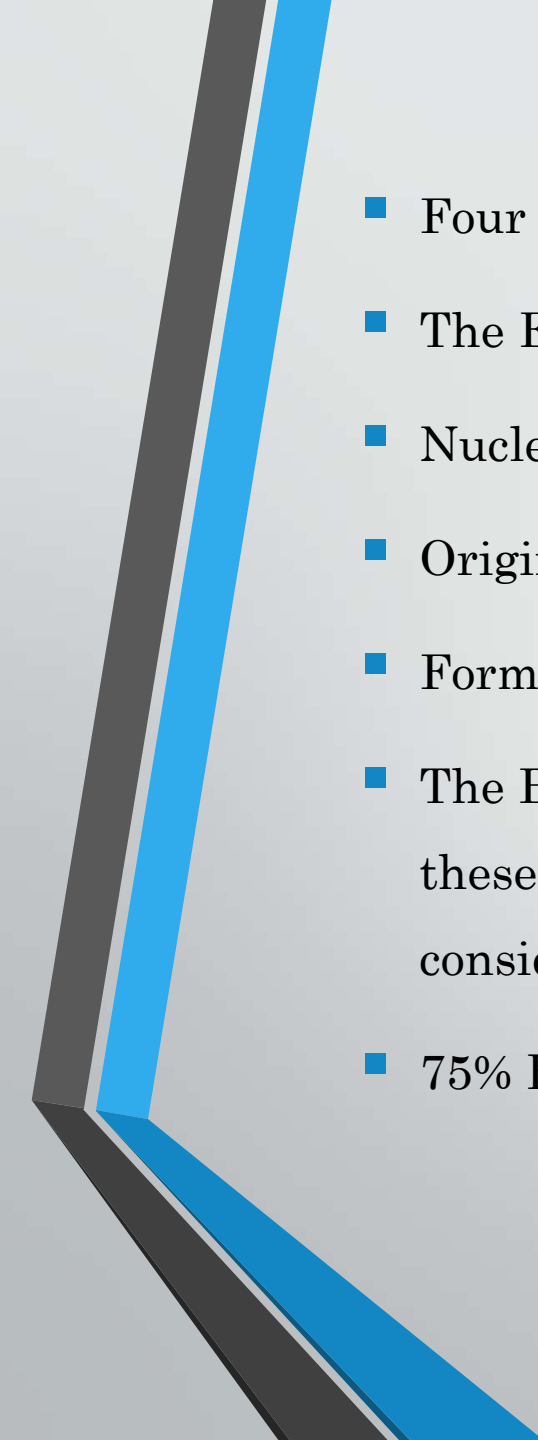
- 
- Einstein's theory of general relativity, announced in 1916, had led to various cosmological models, including Einstein's own model of a static universe.
 - The idea of an expanding universe was demonstrated experimentally in 1929 by Edwin Hubble.
 - In the late 1920's, the astronomer Edwin Hubble first observed that distant galaxies are moving away from us, just as would be expected if the space between galaxies were growing in volume and just as predicted by Einstein's theory of gravity. Since then, astronomers have measured this recession for millions of galaxies.

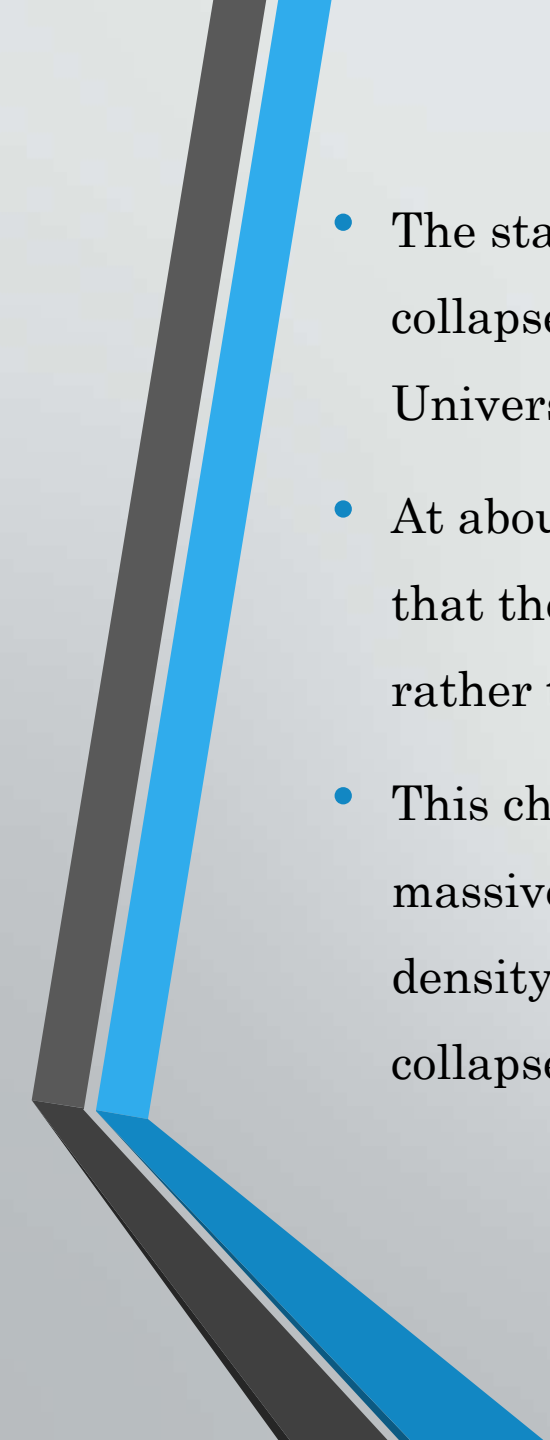
The Big Bang “Theory”

- The Big Bang is actually not a "theory" at all, but rather a scenario or model about the early moments of our universe, for which the evidence is overwhelming.
- But if space and everything with it is expanding now, then the universe must have been much denser in the past. That is, all the matter and energy (such as light) that we observe in the universe would have been compressed into a much smaller space in the past.

- 
- By determining how fast the universe is expanding now, and then "running the movie of the universe" backwards in time, we can determine the age of the universe.
 - The result is that space started expanding 13.77 billion years ago. This number has now been experimentally determined to within 1% accuracy.
 - It's a common misconception that the entire universe began from a point. If the whole universe is infinitely large today (and we don't know yet), then it would have been infinitely large in the past, including during the Big Bang.
 - But any finite chunk of the universe—such as the part of the universe we can observe today—is predicted to have started from an extremely small volume.

- 
- It's also a common misconception that the Big Bang was an "explosion" that took place somewhere in space.
 - But the Big Bang was an expansion of space itself.
 - Every part of space participated in it.
 - For example, the part of space occupied by the Earth, the Sun, and our Milky Way galaxy was once, during the Big Bang, incredibly hot and dense.

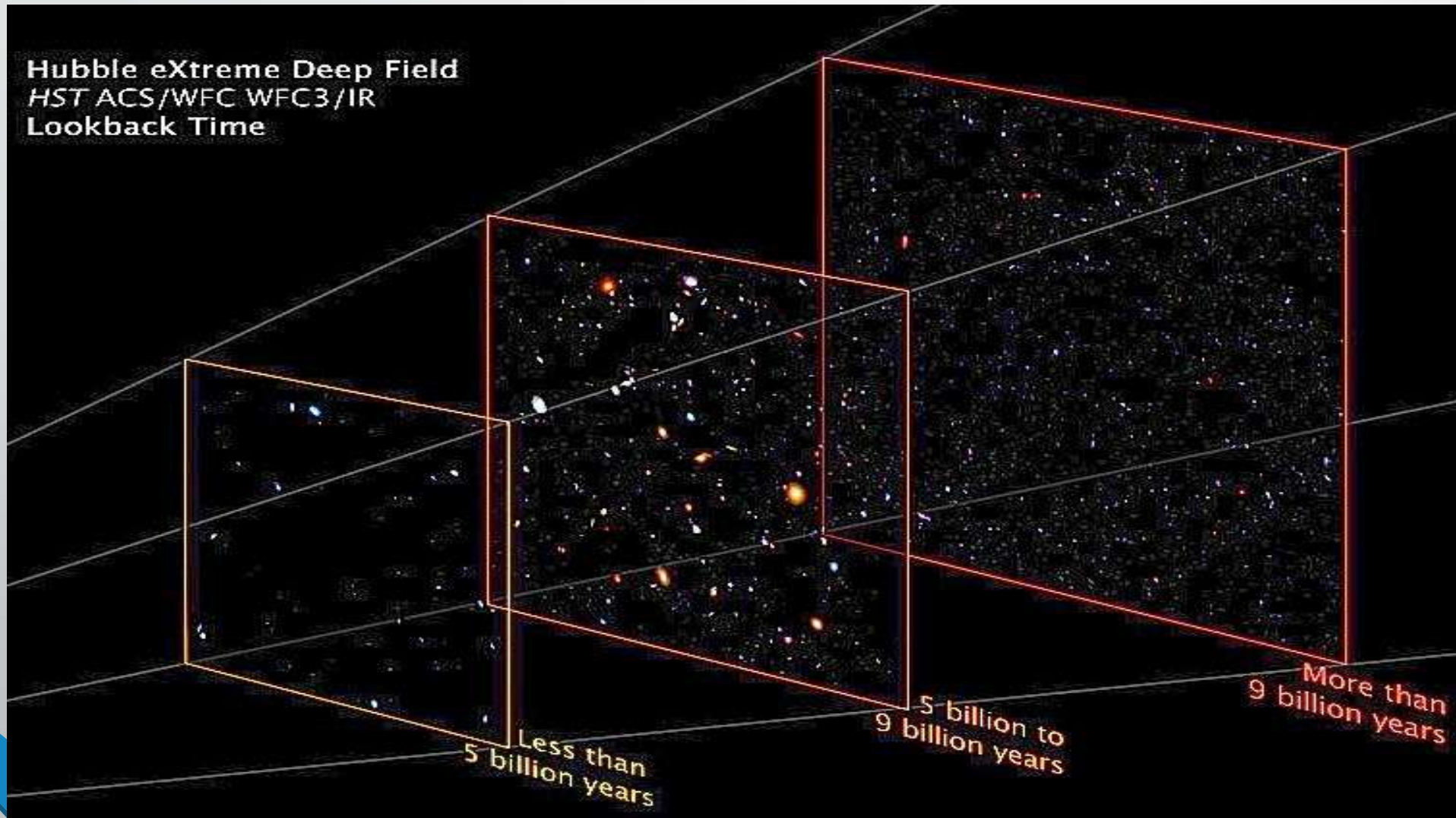
- 
- Four key observational successes of the model are:
 - The Expansion of the Universe
 - Nucleosynthesis of the light elements
 - Origin of the cosmic microwave background radiation
 - Formation of galaxies and large-scale structure
 - The Big Bang model makes accurate and scientifically testable hypotheses in each of these areas, and the remarkable agreement with the observational data gives us considerable confidence in the model.
 - 75% Hydrogen, 25% Helium.

- 
- The standard Hot Big Bang model also provides a framework in which to understand the collapse of matter to form galaxies and other large-scale structures observed in the Universe today.
 - At about 10,000 years after the Big Bang, the temperature had fallen to such an extent that the energy density of the Universe began to be dominated by massive particles, rather than the light and other radiation which had predominated earlier.
 - This change in the form of matter density meant that the gravitational forces between the massive particles could begin to take effect, so that any small perturbations in their density would grow. Thirteen point eight billion years later we see the results of this collapse in the structure and distribution of the galaxies.

The Hubble Space Telescope 1996:



The XDF showing foreground, background, and very far background galaxies



Now
13,700,000,000 YEARS
AFTER BIG BANG



**FORMATION OF
THE SOLAR SYSTEM**
8,700,000,000 YEARS
AFTER BIG BANG

FIRST GALAXIES
1,000,000,000 YEARS
AFTER BIG BANG

**COSMIC MICROWAVE
BACKGROUND**
400,000 YEARS AFTER
BIG BANG

THE DARK AGES

FIRST STARS
400,000,000 YEARS
AFTER BIG BANG

**GALAXY EVOLUTION
CONTINUES...**

DARK ENERGY?

INFLATION

**THE
BIG
BANG**

Cosmos: It's a kind of matter, energy occupied in space

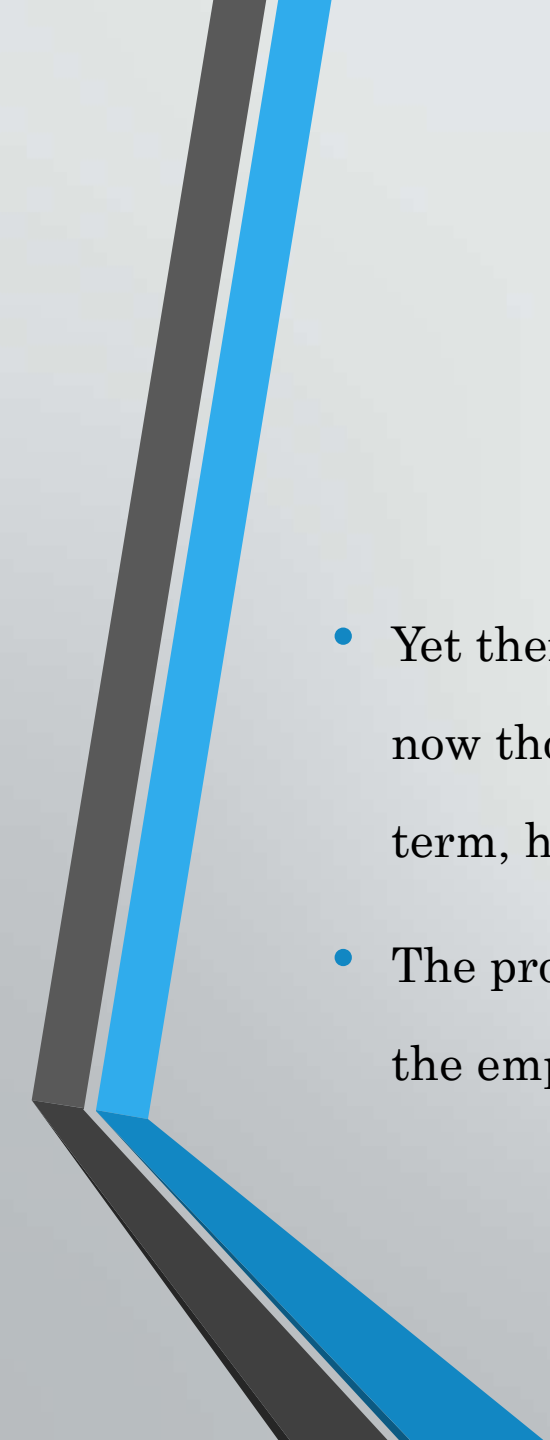


INFINITE UNIVERSE



Vacuum or Dark Energy—a new form of energy driving the cosmic expansion:

- "One proposal for what is driving the current accelerating phase of the universe is the energy of space itself.
- In quantum mechanics even empty space has an energy density in the form of virtual particles that appear and then disappear almost instantaneously.
- On the very small scales where quantum effects become important, even empty space is not really empty.
- Instead virtual particle-antiparticle pairs pop out of the vacuum travel for a short distance and then disappear again on timescales so fleeting that one cannot observe them directly.

- 
- Yet their indirect effects are very important and can be measured. This vacuum energy is now thought of as Einstein's cosmological term. This new concept of the cosmological term, however, is quite different from the equations introduced by Einstein.
 - The problem with this picture, is that all calculations and estimates of the magnitude of the empty-space energy leads to large values.

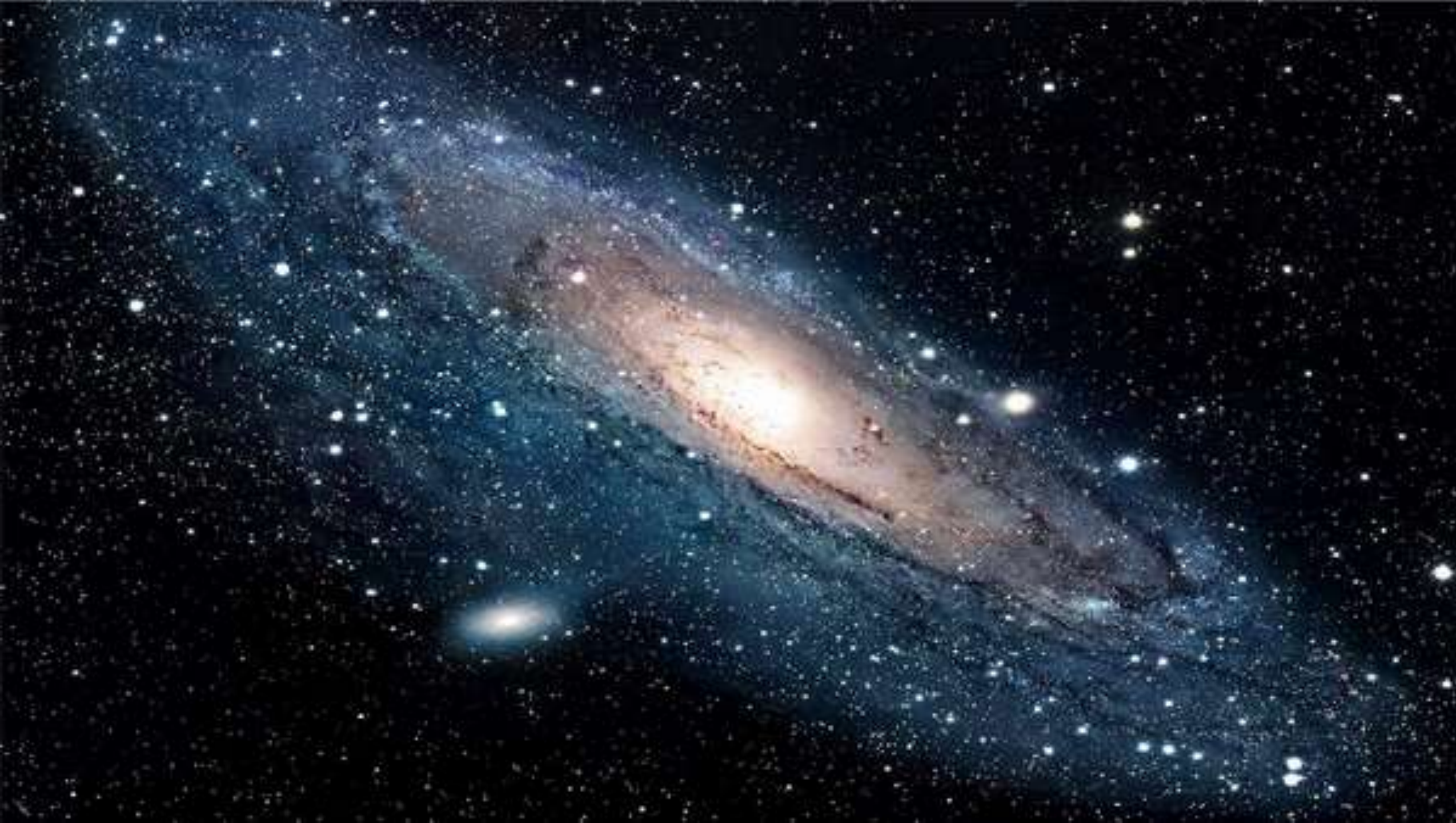
The Milky Way Galaxy:

- Our Sun (a star) and all the planets around it are part of a galaxy known as the Milky Way Galaxy.
- A galaxy is a large group of stars, gas, and dust bound together by gravity. They come in a variety of shapes and sizes.
- The Milky Way is a large barred spiral galaxy. All the stars we see in the night sky are in our own Milky Way Galaxy.
- Our galaxy is called the Milky Way because it appears as a milky band of light in the sky when you see it in a really dark area.

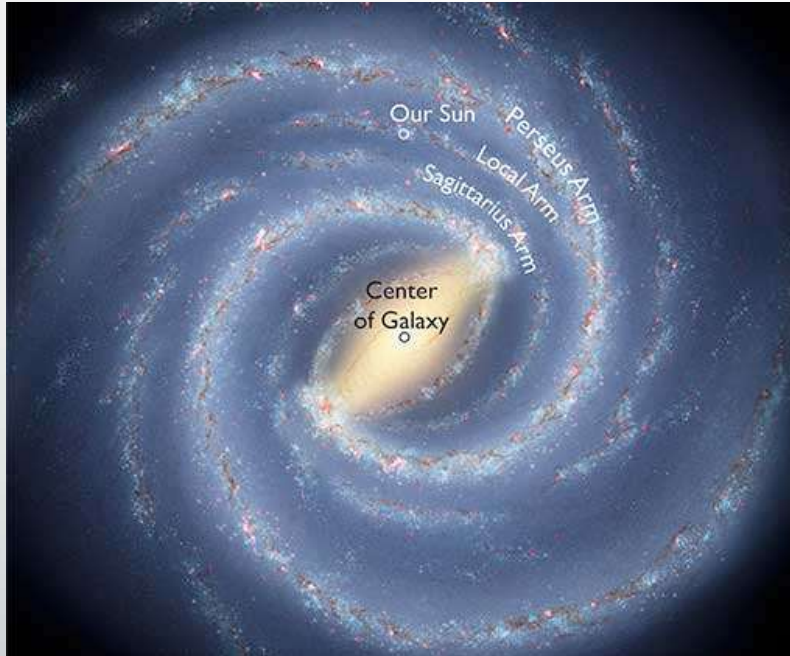
- It is very difficult to count the number of stars in the Milky Way from our position inside the galaxy, so we use Light year.
- Light year is a unit of astronomical distance equivalent to the distance that light travels in one year, which is 9.4607×10^{12} km (nearly 6 million million miles).
- The **Orion Arm** is a minor spiral **arm** of the Milky Way Galaxy that is 3,500 light-years (1,100 parsecs) across and approximately 10,000 light-years (3,100 parsecs) in length, containing the Solar System, including Earth.
- Our best estimates tell us that the Milky Way is made up of approximately 100 billion stars. These stars form a large disk whose diameter is about 100,000 light years.
- Our Solar System is about 25,000 light years away from the center of our galaxy – we live in the suburbs of our galaxy. Just as the Earth goes around the Sun, the Sun goes around the center of the Milky Way. It takes 250 million years for our Sun and the solar system to go all the way around the center of the Milky Way.



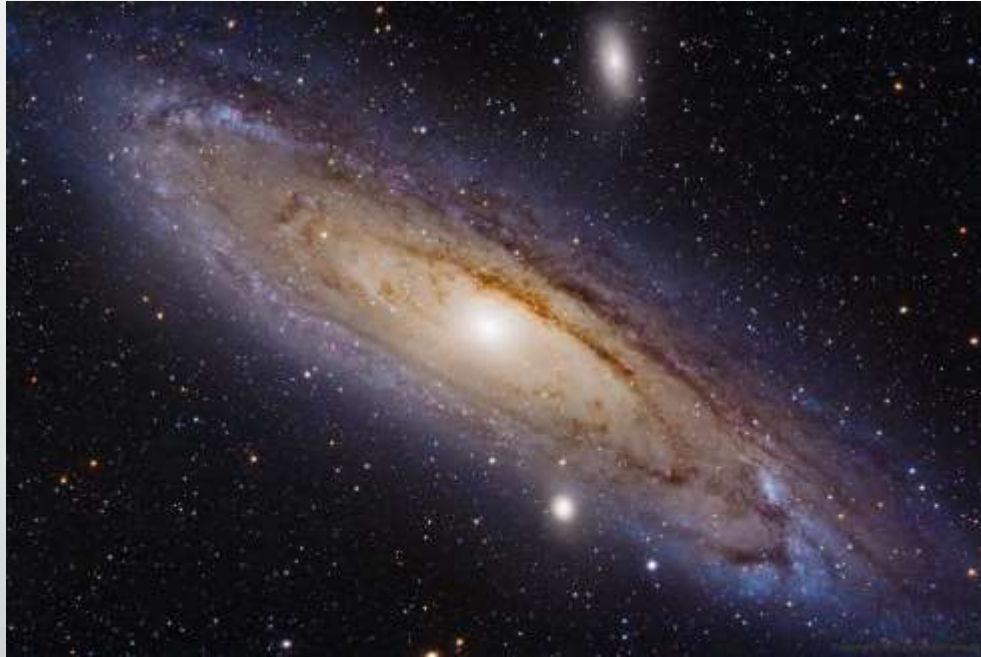
**Galaxy NGC 3370, a spiral galaxy like our
own Milky Way**



Milky way galaxy

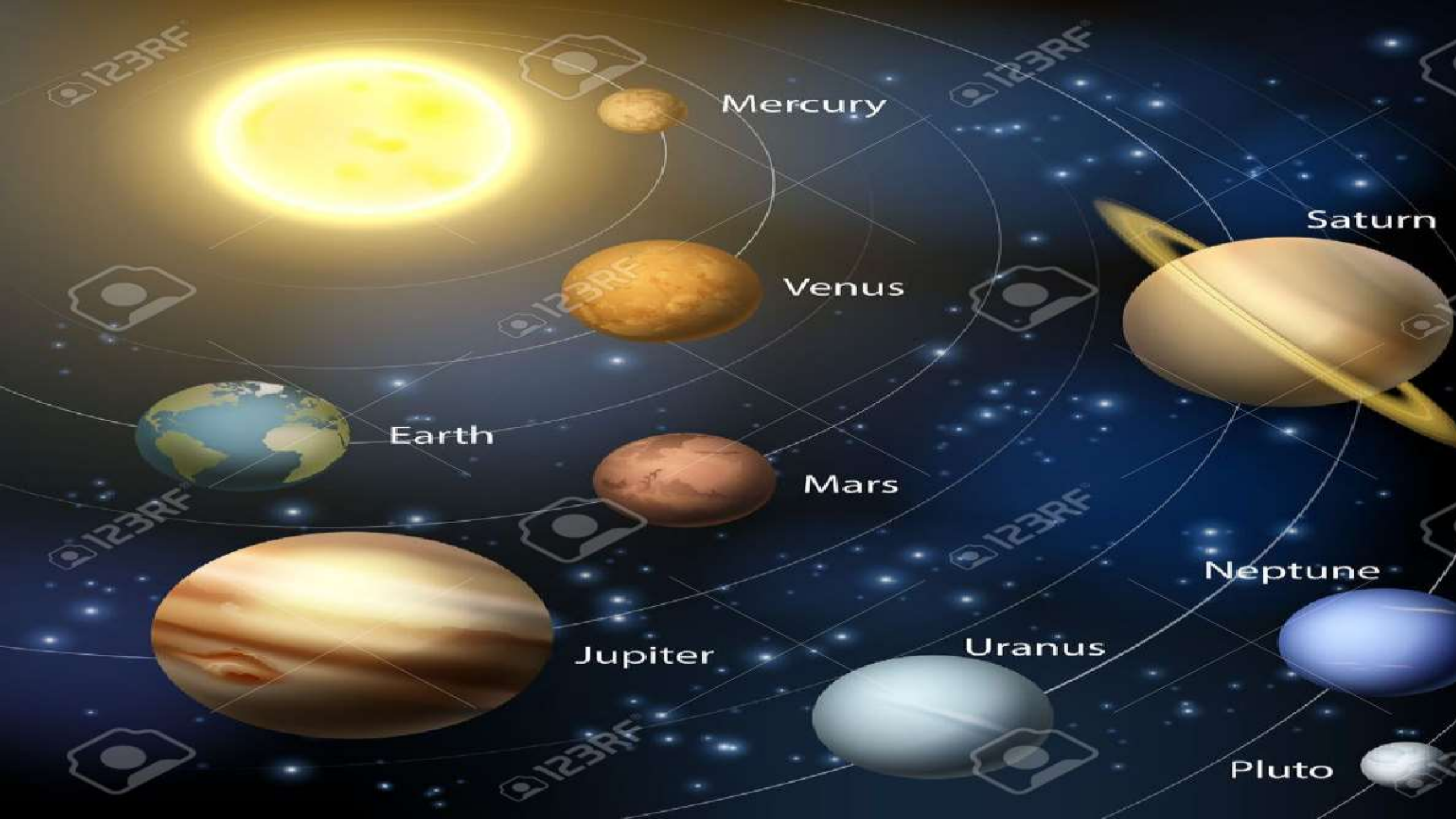


Andromeda Galaxy



The Solar System

Name of the Planet	Distance from Sun	Diameter in Kilometers	Location
Mercury	35.6 million miles	4,880	Solar system
Venus	67.2 million miles	12,102	Solar system
Earth	92.96 million miles	12,756	Solar system
Mars	142.5 million miles	6,794	Solar system
Jupiter	483.6 million miles	142,984	Solar system
Saturn	883.6 million miles	120,536	Solar system
Uranus	1.8 billion miles	51,118	Solar system
Neptune	2.7 billion miles	49,532	Solar system



Mercury

Venus

Earth

Mars

Jupiter

Uranus

Saturn

Neptune

Pluto

Inner planets

- The four **inner planets** have dense, rocky compositions, few or no moons, and no ring system.
- They are composed largely of refractory minerals, such as the silicates—which form their crusts and mantles—and metals, such as iron and nickel, which form their cores.
- Three of the four inner planets (Venus, Earth and Mars) have atmospheres substantial enough to generate weather; all have impact craters and tectonic surface features, such as rift valleys and volcanoes.

Mercury

- Mercury (0.4 AU from the Sun) is the closest planet to the Sun. {1AU=1.496E+8Km}
- Mercury is the smallest planet in the Solar System.
- Mercury has no natural satellites.
- Its relatively large iron core and thin mantle have not yet been adequately explained. Hypotheses include that its outer layers were stripped off by a giant impact, or that it was prevented from fully accreting by the young Sun's energy

Venus

- Venus (0.7 AU from the Sun) is close in size to Earth and, like Earth, has a thick silicate mantle around an iron core, a substantial atmosphere, and evidence of internal geological activity.
- It is much drier than Earth, and its atmosphere is ninety times as dense.
- Venus has no natural satellites.
- It is the hottest planet, with surface temperatures over 400 °C , most likely due to the amount of greenhouse gases in the atmosphere. No definitive evidence of current geological activity has been detected on Venus, but it has no magnetic field that would prevent depletion of its substantial atmosphere, which suggests that its atmosphere is being replenished by volcanic eruptions.

Earth:

- Earth (1 AU from the Sun) is the largest and densest of the inner planets, the only one known to have current geological activity, and the only place where life is known to exist. $M_{\oplus} = 5.9722 \times 10^{24} \text{ kg}$.
- Its liquid hydrosphere is unique among the terrestrial planets, and it is the only planet where plate tectonics has been observed.
- Earth's atmosphere is radically different from those of the other planets, having been altered by the presence of life to contain 21% free oxygen.
- It has one natural satellite, the Moon, the only large satellite of a terrestrial planet in the Solar System.

Mars:

- Mars (1.5 AU from the Sun) is smaller than Earth and Venus. It has an atmosphere of mostly carbon dioxide with a surface pressure of 6.1 millibars (roughly 0.6% of that of Earth).
- Its surface, peppered with vast volcanoes, such as Olympus Mons, and rift valleys, such as Valles Marineris, shows geological activity that may have persisted until as recently as 2 million years ago.
- Its red colour comes from iron oxide (rust) in its soil.
- Mars has two tiny natural satellites (Deimos and Phobos) thought to be either captured asteroids, or ejected debris from a massive impact early in Mars's history

Outer Solar System:

- The outer region of the Solar System is home to the giant planets and their large moons.
- The centaurs and many short-period comets also orbit in this region. Due to their greater distance from the Sun, the solid objects in the outer Solar System contain a higher proportion of volatiles, such as water, ammonia, and methane than those of the inner Solar System because the lower temperatures allow these compounds to remain solid.

Jupiter:

- Jupiter (5.2 AU), at $318 M_{\oplus}$, is 2.5 times the mass of all the other planets put together. It is composed largely of hydrogen and helium.
- Jupiter's strong internal heat creates semi-permanent features in its atmosphere, such as cloud bands and the Great Red Spot.
- Jupiter has 79 known satellites.
- The four largest, Ganymede, Callisto, Io, and Europa, show similarities to the terrestrial planets, such as volcanism and internal heating.
- Ganymede, the largest satellite in the Solar System, is larger than Mercury.

Saturn:

- Saturn (9.5 AU), distinguished by its extensive ring system, has several similarities to Jupiter, such as its atmospheric composition and magnetosphere.
- Although Saturn has 60% of Jupiter's volume, it is less than a third as massive, at $95 M_{\oplus}$.
- Saturn is the only planet of the Solar System that is less dense than water. The rings of Saturn are made up of small ice and rock particles.
- Saturn has 82 confirmed satellites composed largely of ice.
- Two of these, Titan and Enceladus, show signs of geological activity.
- Titan, the second-largest moon in the Solar System, is larger than Mercury and the only satellite in the Solar System with a substantial atmosphere.

Uranus:

- Uranus (19.2 AU), at $14 M_{\oplus}$, is the lightest of the outer planets. Uniquely among the planets, it orbits the Sun on its side; its axial tilt is over ninety degrees to the ecliptic. It has a much colder core than the other giant planets and radiates very little heat into space. Uranus has 27 known satellites, the largest ones being Titania, Oberon, Umbriel, Ariel, and Miranda.

Neptune

- Neptune (30.1 AU), though slightly smaller than Uranus, is more massive ($17 M_{\oplus}$) and hence more dense. It radiates more internal heat, but not as much as Jupiter or Saturn.
- Neptune has 14 known satellites. The largest, Triton, is geologically active, with geysers of liquid nitrogen.
- Triton is the only large satellite with a retrograde orbit.
- Neptune is accompanied in its orbit by several minor planets, termed Neptune trojans, that are in 1:1 resonance with it.

Pluto

- The dwarf planet Pluto (39 AU average) is the largest known object in the Kuiper belt.
- When discovered in 1930, it was considered to be the ninth planet; this changed in 2006 with the adoption of a formal definition of planet. Pluto has a relatively eccentric orbit inclined 17 degrees to the ecliptic plane and ranging from 29.7 AU from the Sun at perihelion (within the orbit of Neptune) to 49.5 AU at aphelion.
- Pluto has a 3:2 resonance with Neptune, meaning that Pluto orbits twice round the Sun for every three Neptunian orbits. Kuiper belt objects whose orbits share this resonance are called plutinos.
- Charon, the largest of Pluto's moons, is sometimes described as part of a binary system with Pluto, as the two bodies orbit a barycentre of gravity above their surfaces (i.e. they appear to "orbit each other").
- Beyond Charon, four much smaller moons, Styx, Nix, Kerberos, and Hydra, orbit within the system.

Origin of Stars

- One scenario envisioning the creation of the earliest stars suggests that dark matter particles are very light and can zip through space more quickly.
- These warm dark matter models predict that dark matter formed long filamentary structures along which stars appeared like pearls on a string.
- In this simulation, a gas filament condenses and then fragments to form the first stars.
- **Stars** are classified by their spectra (the elements that they absorb) and their temperature.

SUN IS A STAR

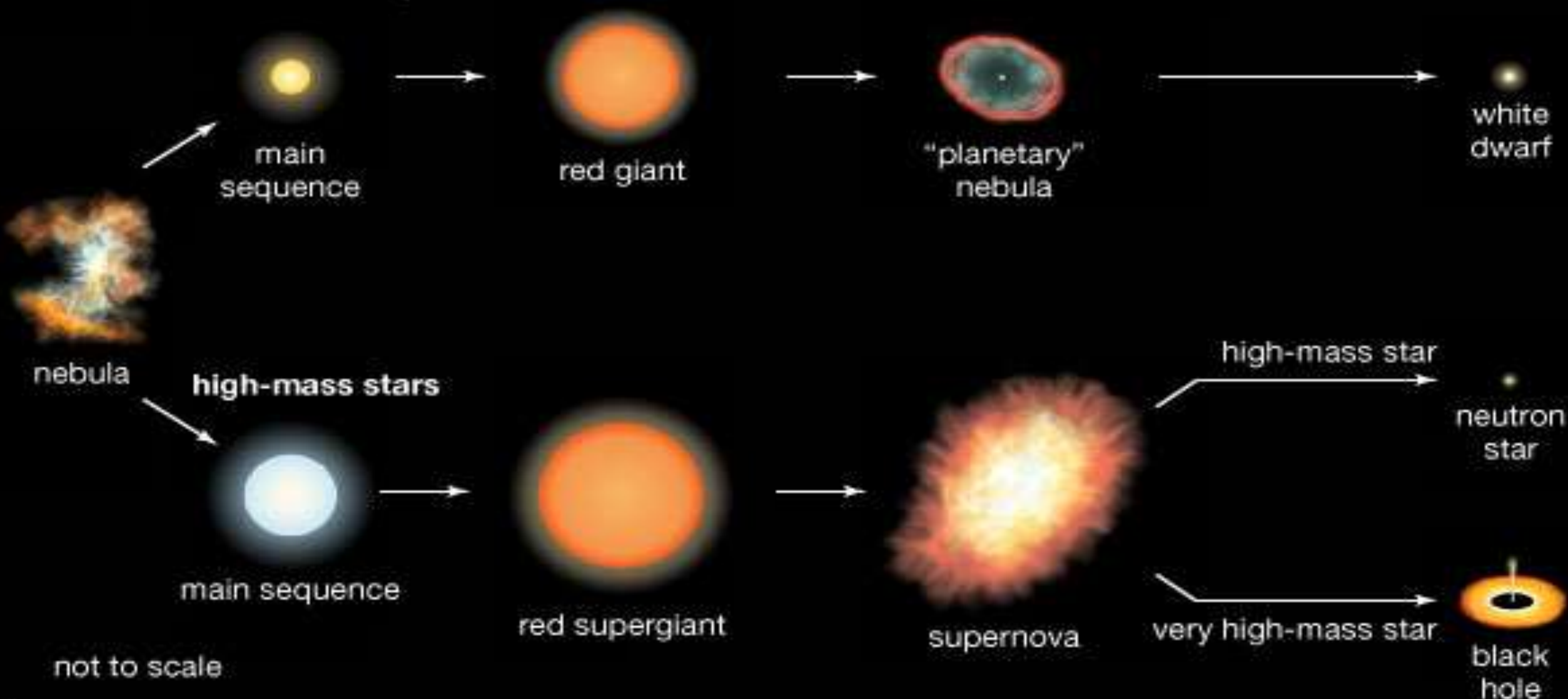
- The **Sun** is a Golden Yellow star, and the heart of our **solar** system. Its gravity holds everything together in the solar system from the biggest planets to the smallest particles in its orbit.
- Compared with the billions of other stars in the universe, there are some stars bigger than sun. But, the **sun** is a powerful star in our solar system, and it is the centre of attraction. For Earth and the other planets that revolve around it, Sun holds the solar system together; providing light, heat, and energy to all the living species on Earth and also generates space weather.
- One million Earths could fit inside the **Sun**.
- The **Sun** contains 99.86% of the mass in the Solar System. ...
- The **Sun** is an almost perfect sphere.
- Our body creates **vitamin D** from direct sunlight on our skin when we're outdoors.

How stars are formed

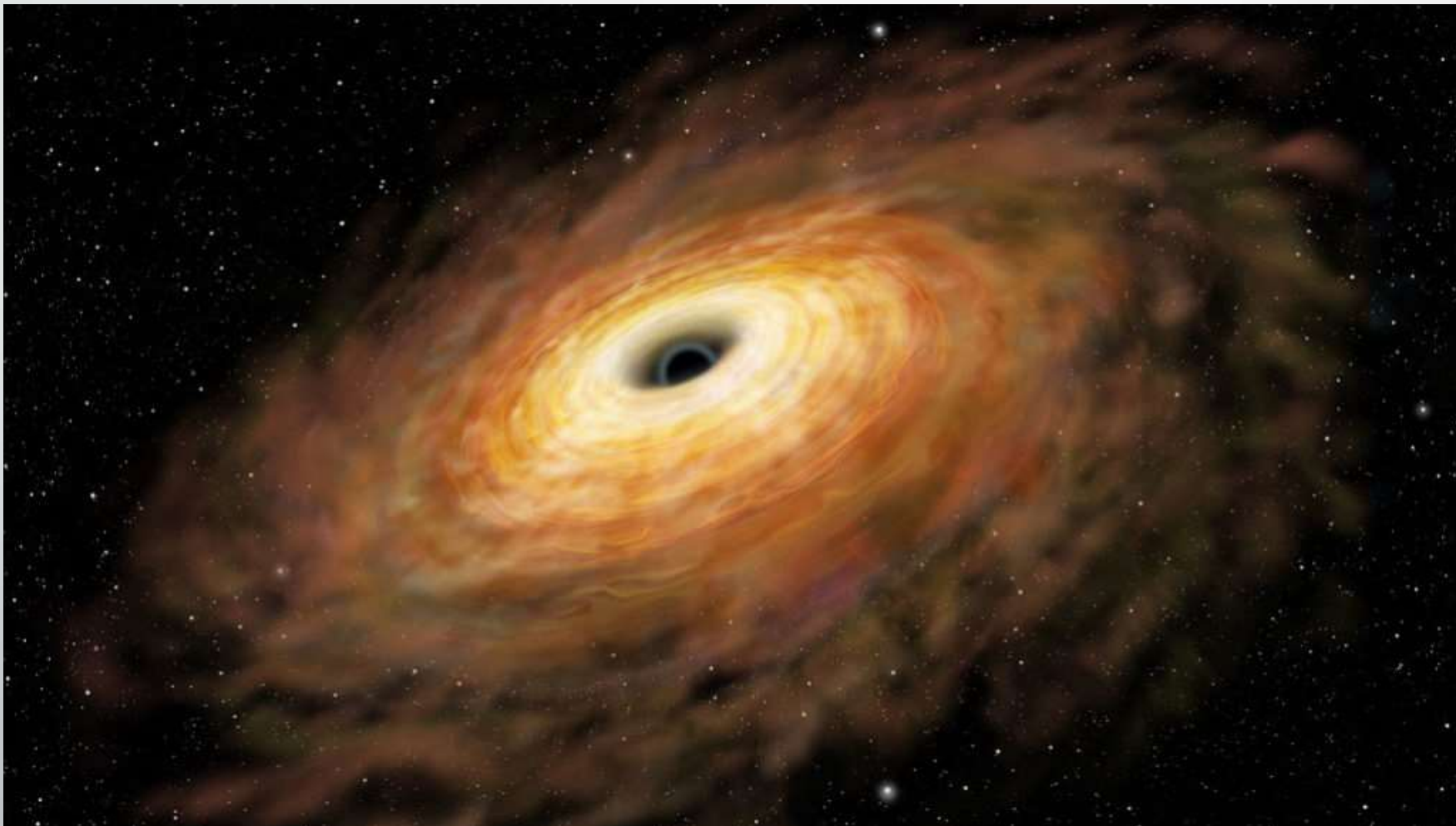
- **Nebulae** are made of dust and gases mostly hydrogen and helium. The collapse causes the material at the center of the cloud to heat up and this hot core **is the beginning** of a **star**.
- **Stars form** from an accumulation of gas and dust, which collapses due to gravity and starts to **form stars**. The process of star **formation** takes around a million years from the time the initial gas cloud starts to collapse until the star is created and shines like the Sun.
- **Stars** are born, **live**, and die. This process is called the "life cycle of a **star**". Most of the time a **star** shines, it is in a stage of its life cycle called the main sequence.

Stellar evolution

low- and medium-mass stars
(including the Sun)



Black Hole:

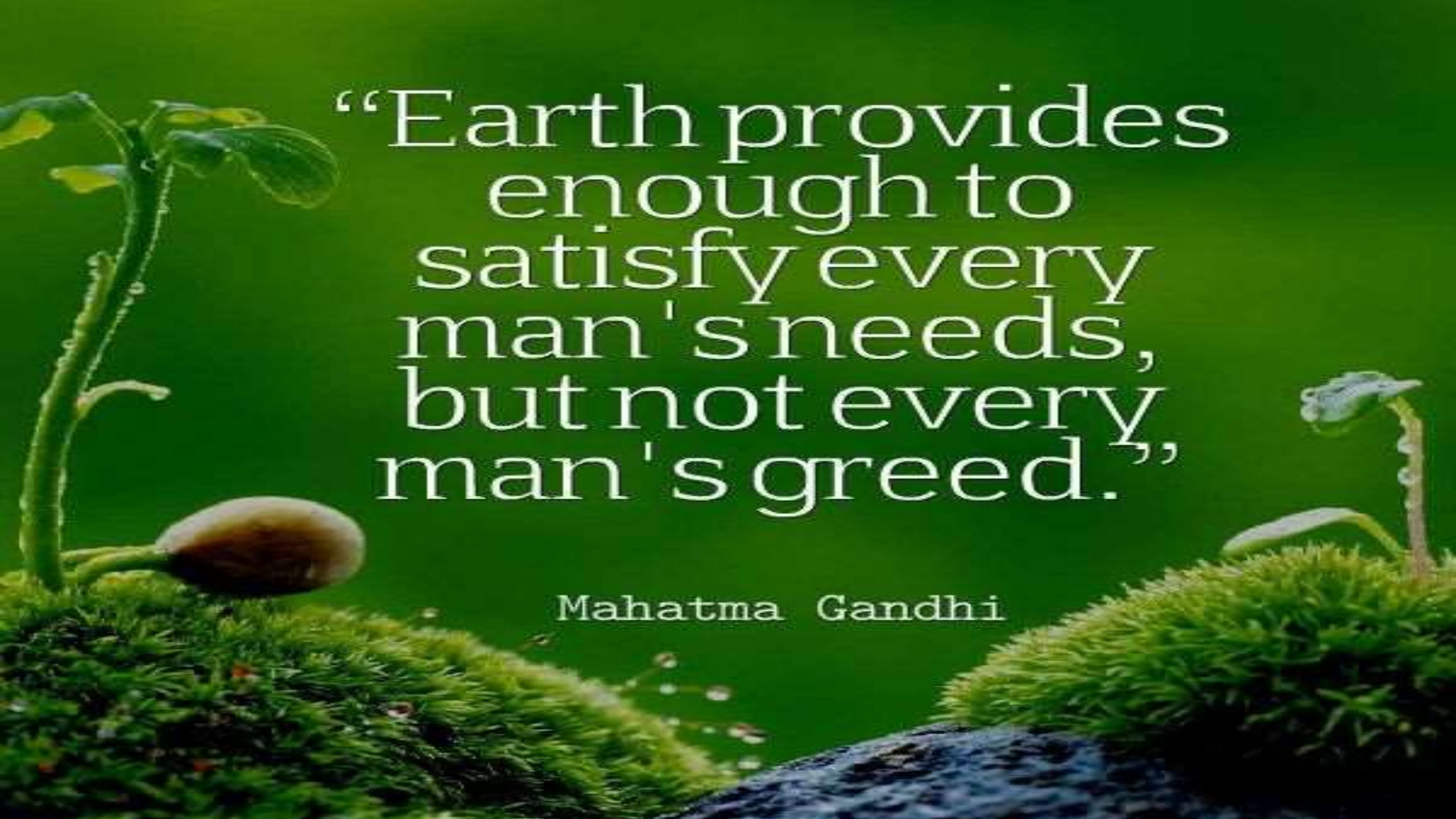


Significance of stars:

- **Uses of Stars:**
- Plenty of stars are present in the universe for long period of time due to their significance effect they sustain today also. **VY Canis Majoris**, is up to 2,100 times the size of the Sun and UY Scuti, a hypergiant with a radius around 1,700 times **larger than the sun**.
- The main property of the star is to give brightness. And it contains lot many significance that we have already studied with our own god particle sun.
- But, Some stars forms like a Jwala(fire) stars which will be harmful to the neighbouring stars, planets, and so on in such conditions the stars shatter into pieces,
- Some stars forms like a black hole, where the star converts into a huge wind destroying even a sparkle of light, we consider it as death of star.

OUR PLANET EARTH





“Earth provides
enough to
satisfy every
man's needs,
but not every
man's greed.”

Mahatma Gandhi







HEAVEN ON EARTH

































Earth Belongs to Everyone



HELL ON EARTH













What happens when a nuclear bomb explodes?

Immediate aftermath

The degree of damage depends upon the distance from the center of the blast. Within a ground zero, death, injuries and human suffering were likely to be the greatest. Beyond the urban devastation, forests, agricultural fields and crops were destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

Within the immediate urban area, destruction was caused by heat, radiation, and flying debris. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

Long-term effects

The long-term effects of the atomic bomb on Hiroshima were not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

"There is no doubt that, if the people of the world were more fully aware of the inherent danger of nuclear weapons and the consequences of their use, they would reject them, and not permit their continued production or acquisition on their behalf by their governments, even for an alleged need for self-defense."

—Statement of the United Nations Committee on the Disarmament of Atomic Weapons

Health effects

Health effects of the atomic bomb were not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

Health effects of the atomic bomb were not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

Health effects of the atomic bomb were not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

Environmental damage

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

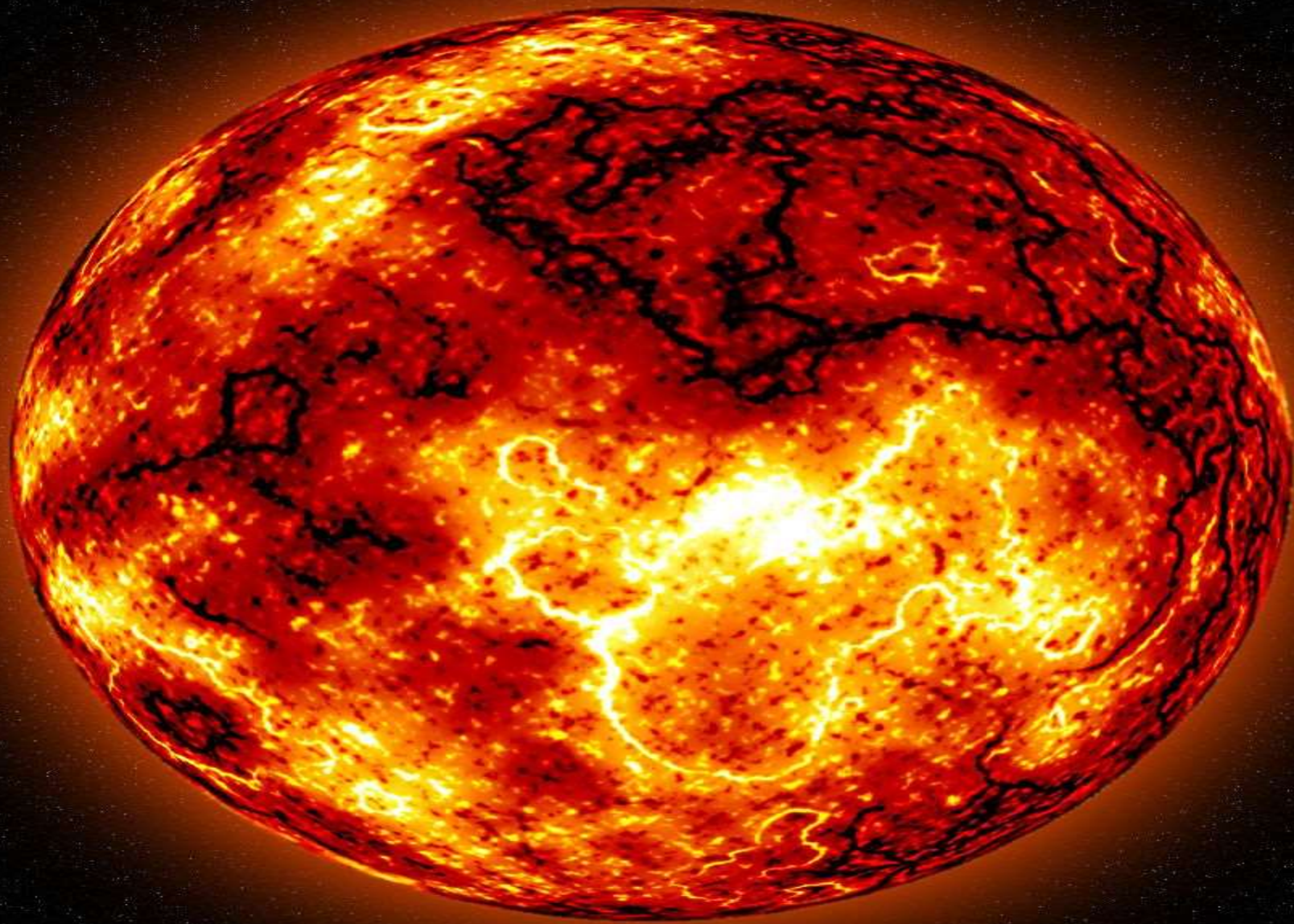
The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

The environmental damage caused by the atomic bomb was not limited to the immediate aftermath. The city of Hiroshima was almost completely destroyed. The devastation was not limited to the ground. The atomic bomb destroyed the city of Hiroshima, Japan, on August 6, 1945.

THIS CAN'T BE *GOOD*...
RIGHT?!

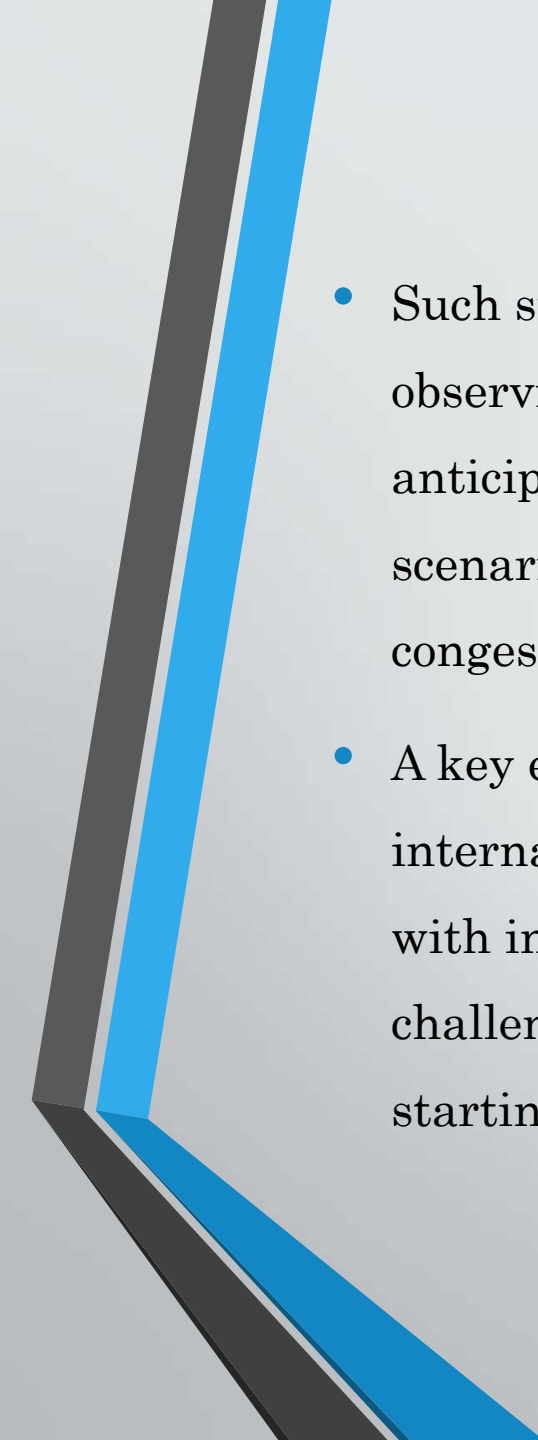






Space is getting crowded: the challenge of space traffic management:

- More than 60 years ago the first spacecraft was launched into the orbit, the number of satellites orbiting the Earth has significantly increased
- The dependency of everyday technological systems are safe or not with continuous satellite operations is not surveyed.
- As the number of launches is expected to further boost in the near future, some orbital highways might soon be on the verge of congestion and the risk of collisions among objects unsustainable.

- 
- Such situations can be handled by a complex system including several elements: observing the sky to monitor and track satellites in orbit, predicting their trajectories, anticipating the risk of collision and re-entry on the ground and acting on hazardous scenarios, as well as adapting future missions and spacecraft design to the risk of congested orbits.
 - A key element relies on finding an appropriate solution for the handling of space traffic at international level, balancing on one side safety and sustainability in space and on earth with innovation and development. This represents one of the most dynamic and intricate challenges of the years to come as diverse scientific, legal and governance questions are starting to be investigated and are yet to be answered

- 
- Let us make our Earth –The Heaven
 - The Universe is eternal in time and infinite in space. There is no "outside of" the Universe.

THANK YOU